

Newton's method

Suppose p is a root of f and p_0 is sufficiently close to p , then

$$f(p) = f(p_0) + f'(p_0)(p - p_0) + \frac{1}{2}f''(\xi(p))(p - p_0)^2$$

for some $\xi(p)$ between p_0 and p .

Since $f(p) = 0$, and $(p - p_0)^2$ is close to 0, we have

$$0 \approx f(p_0) + f'(p_0)(p - p_0)$$

Therefore (assume $f'(p_0) \neq 0$),

$$p \approx p_0 - \frac{f(p_0)}{f'(p_0)} =: p_1$$

So p_1 is our guess for p now!

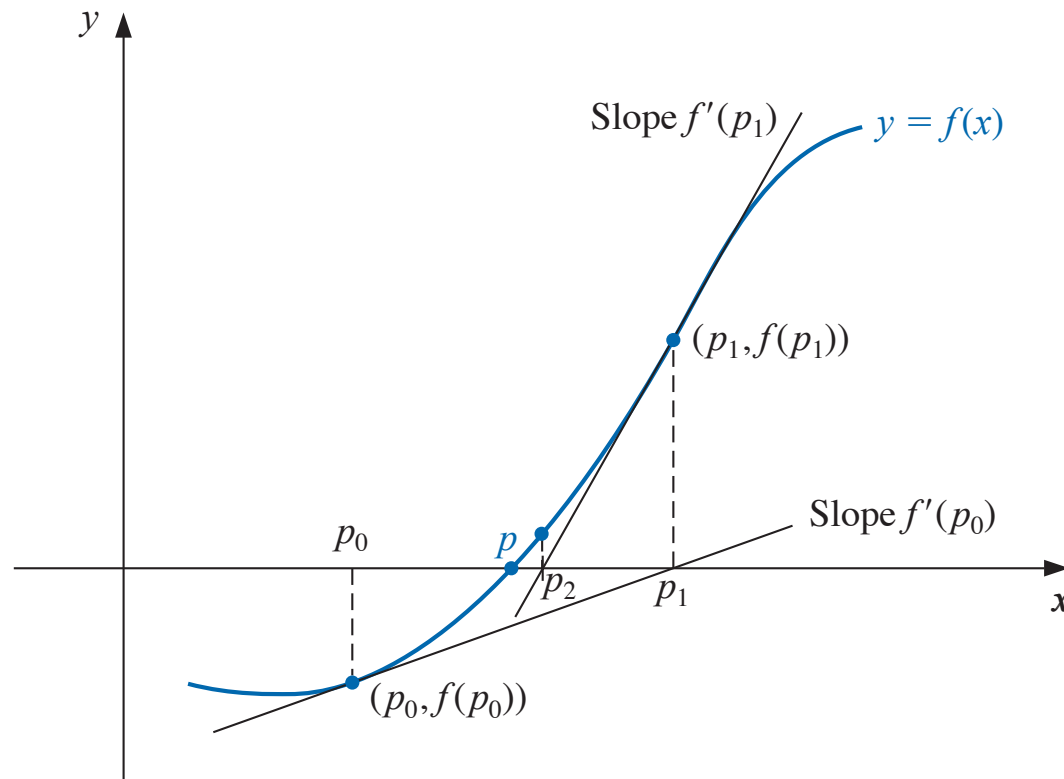
Newton's method

Newton's method:

Start from initial guess p_0 (close to the FP p), and iterate:

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}$$

Then we hope $p_n \rightarrow p$ quickly.



Newton's method

Newton's method

- ▶ **Input.** Initial guess p_0 , ϵ_{tol} , N_{max} . Set $N = 1$.
- ▶ While $N \leq N_{\text{max}}$, do:
 1. Set $p = p_0 - \frac{f(p_0)}{f'(p_0)}$ (compute p_n using p_{n-1})
 2. If $|p - p_0| < \epsilon_{\text{tol}}$, STOP
 3. Set $N = N + 1$
 4. Set $p_0 = p$ (update p_{n-1} using p_n for next iteration)
- ▶ **Output.** Approximate solution p . If $N \geq N_{\text{max}}$, print("Max iteration reached").

Newton's method

Newton's method is equivalent to fixed point iteration algorithm with

$$g(x) := x - \frac{f(x)}{f'(x)}$$

So p is a FP of g iff $f(p) = 0$.

Convergence of Newton's method

Theorem

If $f \in C^2[a, b]$ and $\exists p \in (a, b)$ such that $f(p) = 0$ and $f'(p) \neq 0$, then $\exists \delta > 0$ such that Newton's method convergent starting from any $p_0 \in (p - \delta, p + \delta)$.

Proof.

Hint: Check $g'(x)$:

$$g'(x) = 1 - \frac{(f')^2 - ff''}{(f')^2} = \frac{f(x)f''(x)}{(f'(x))^2}$$

$f \in C^2$, $f(p) = 0$, $f'(p) \neq 0$ together imply $\exists \delta > 0$ s.t. $|g'(x)| < 1$ for $x \in (p_0 - \delta, p_0 + \delta)$. □

Secant method

A problem with Newton's method is that $f'(x)$ may not be easy to calculate, so we approximate $f'(p_{n-1})$ in the Newton's method by

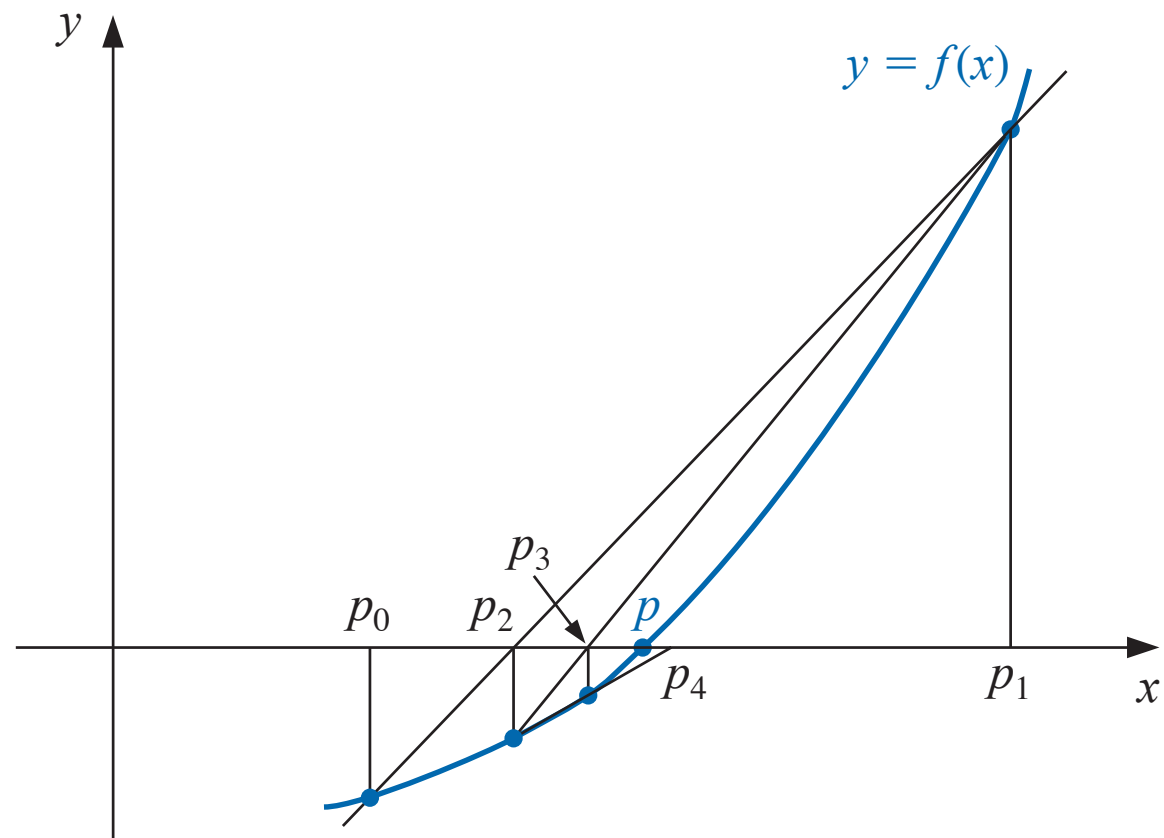
$$f'(p_{n-1}) \approx \frac{f(p_{n-1}) - f(p_{n-2})}{p_{n-1} - p_{n-2}}$$

After simple algebra, we get the **secant method**:

$$\begin{aligned} p_n &= p_{n-1} - \frac{f(p_{n-1})(p_{n-1} - p_{n-2})}{f(p_{n-1}) - f(p_{n-2})} \\ &= \frac{p_{n-2}f(p_{n-1}) - p_{n-1}f(p_{n-2})}{f(p_{n-1}) - f(p_{n-2})} \end{aligned}$$

Secant method

Illustration of the secant method:



Secant method

Secant method

- ▶ **Input.** Initial guess p_0, p_1 , $q_0 = f(p_0)$, $q_1 = f(p_1)$, ϵ_{tol} , N_{max} . Set $N = 1$.
- ▶ While $N \leq N_{\text{max}}$, do:
 1. Set $p = p_1 - \frac{q_1(p_1 - p_0)}{q_1 - q_0} = \frac{p_0 q_1 - p_1 q_0}{q_1 - q_0}$ (compute p_n using p_{n-1}, p_{n-2})
 2. If $|p - p_1| < \epsilon_{\text{tol}}$, STOP
 3. Set $N = N + 1$
 4. Set $p_1 = p$, $q_1 = f(p)$, $p_0 = p_1$, $q_0 = q_1$ (update p_{n-1}, p_{n-2} for next iteration)
- ▶ **Output.** Approximate solution p . If $N \geq N_{\text{max}}$, print(“Max iteration reached”).